NARAYANA IIT ACADEMY INDIA

Sec: Sr. IIT **Date: 27-02-24**

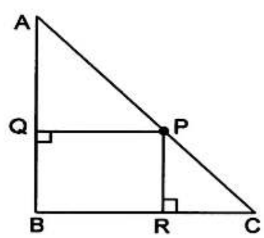
Daily Practices Sheet - 06

MATHS

1. If $x\cos\alpha + y\sin\alpha = x\cos\beta + y\sin\beta = a\left(0<\alpha,\beta<\dfrac{\pi}{2}\right)$, then the value of $\cos(\alpha+\beta)$ is

1) $\dfrac{4axy}{x^2+y^2}$ 2) $\dfrac{2a^2-x^2-y^2}{x^2+y^2}$ 3) $\dfrac{x^2-y^2}{x^2+y^2}$ 4) $\dfrac{a^2-x^2}{a^2-y^2}$

2. ABC is an isosceles right angled triangle with AB = BC = 1. If P is any point on AC, then min{max{area APQ, (area PQBR), area PRC}}



1) $\dfrac{2}{9}$ 2) $\dfrac{1}{9}$ 3) $\dfrac{1}{3}$ 4) None of these

3. Number of ordered pair (x, y) satisfying $x^2+1=y$ and $y^2+1=x$ is_________ $(x, y \in R)$

1) 0 2) 1 3) 2 4) 4

4. If a, b, c and d are distinct integers such that a + b + c +d = 17, then the real roots of $(x-a)(x-b)(x-c)(x-d)=4$ are

1) all rational 2) all integers
3) irrational 4) nothing can be said

5. Let $f(x)=x+\sin x$, then $\int_a^1 f^{-1}(x)\,dx$ is equal to, where

$a=\lim_{x\to 0^+}\left[\sec^{-1}\left(\dfrac{1}{x}\right)-\sec^{-1}\left(-\dfrac{1}{x}\right)\right]$, (where [.] denotes the greatest integer function)

1) $-2+\sin 1$ 2) 0 3) $2(1+\sin 1)$ 4) 3

6. $\lim\limits_{n\to\infty}\int\limits_0^1 \frac{nx^{n-1}}{1+x^2}dx =$

1) 0 2) 1 3) 2 4) $\frac{1}{2}$

7. If a, b, c are non zero real numbers and the equation $ax^2+bx+c+i=0$ has purely imaginary roots then

1) $a=bc$ 2) $a=b^2c$ 3) $a=\sqrt{bc}$ if $b>0$ 4) None of these

8. Let u, v, w, z be complex numbers such that $|u|<1, |v|=1$ and $w=\frac{v(u-z)}{\bar{u}z-1}$, then

1) $|w|\le 1 \Rightarrow |z|<1$ 2) $|w|\ge 1 \Rightarrow |z|\ge 1$ 3) $|w|\le 1 \Rightarrow |z|\ge 1$ 4) $|w|\ge 1 \Rightarrow |z|\le 1$

9. P and Q are any two points on the circle $x^2+y^2=4$ such that PQ is a diameter. If α and β are the lengths of perpendiculars from P and Q on x + y = 1, then the maximum value of $\alpha\beta$ is

1) $\frac{1}{2}$ 2) $\frac{7}{2}$ 3) 1 4) 2

10. There is a point P(3, 0) inside the circle $x^2+y^2=r^2$, a chord AB of the circle passes through point P such that AP = 2 and BP = 8, then the radius of the circle is

1) 3 2) 4 3) 5 4) $\sqrt{3+4+5}$

11. The constant term in the expansion of $\left(x+x^2+\frac{1}{x}+\frac{1}{x^2}\right)^{15}$ is

1) $\left({}^{15}C_0\right)^2$ 2) ${}^{30}C_{15}$ 3) $2.{}^{15}C_0$ 4) $\sum\limits_{r=0}^{5}\left({}^{15}C_{5+r}\right)\left({}^{15}C_{3r}\right)$

12. Let S be the set of 6 – digit numbers $a_1 a_2 a_3 a_4 a_5 a_6$ (all digits distinct) where $a_1>a_2>a_3>a_4<a_5<a_6$. Then n(S) is equal to

1) 210 2) 2100 3) 4200 4) 420

13. The angle of elevation of the top of the tower observed from each of the three points A,B,C on the ground, forming a triangle is the same angle α. If R is the circum radius of the triangle ABC, then the height of the tower is

1) $R\sin\alpha$ 2) $R\cos\alpha$ 3) $R\cot\alpha$ 4) $R\tan\alpha$

14. The number of ordered pairs (x, y) satisfying the system of equations given by $\sin x + \sin y = \sin(x+y)$ and $|x|+|y|=1$ is

1) 2 2) 4 3) 6 4) None of these

15. The number of real solutions of the equation $\log_7(x+2)=6-x$ is

1) 0 2) 1 3) 2 4) 3

16. The set of values of x for which $\left(x^2+x+1\right)^x<1$ is

1) $(-\infty,-1)$ 2) $(-\infty,1)$ 3) $(-\infty,0)$ 4) None of these

17. If $f(x)=\int_0^x\left(e^t+3\right)dt$, then which of the following is always true $\forall x \in R$

1) $f(x-3)-f(x-1)\le 11$
2) $f(x+1)-f(x-1)\le 11$
3) $f(x+3)-f(x-5)\ge 24$
4) $f(x)+f(x+1)<-21$

18. The equation of curve passing through (1, 1) in which the subtangent is always bisected at the origin is

1) $y^2=x$ 2) $2x^2-y=1$ 3) $x^2+y^2=2$ 4) $x+y=2$

19. Three friends whose ages form a G.P. divide a certain sum of money in proportion to their ages. If they do that three years later, when the youngest is half the age of the oldest then he will receive Rs. 150/- more than he gets now, and the middle friend will get Rs. 15 more than now, then the age of the youngest friend is

1) 27 2) 18 3) 14 4) 12

20. The first two terms of an H.P. are $\frac{2}{5}$ and $\frac{12}{23}$. The value of the largest term of the H.P. is

1) $\frac{72}{73}$ 2) 6 3) $\frac{1}{6}$ 4) None of these

21. The sides of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are

1) $a\sqrt{2}, b\sqrt{2}$ 2) $\sqrt{a},\sqrt{b}$ 3) a, b 4) $\frac{a}{\sqrt{2}},\frac{b}{\sqrt{2}}$

22. If the difference of the eccentric angles of two point P and Q on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{\pi}{2}$ and R is the point of intersection of tangents at P and Q, then POQR is necessarily (where O is the origin) a

1) rectangle
2) parallelogram
3) cyclic quadrilateral
4) square

23. If $I_1 = \lim_{x\to\infty}\left(\tan^{-1}\pi x - \tan^{-1}x\right)\cos x$ and $I_2 = \lim_{x\to-\infty}\left(\tan^{-1}\pi x - \tan^{-1}x\right)\cos x$, then $\left(I_1, I_2\right)$ is

1) (0, 0) 2) (0, 1) 3) (1, 0) 4) None of these

24. Let a be a real number and $\vec{\alpha} = \hat{i} + 2\hat{j}, \vec{\beta} = 2\hat{i} + a\hat{j} + 10\hat{k}$, $\vec{\gamma} = 12\hat{i} + 2a\hat{j} + a\hat{k}$ be three vectors, then $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are

1) linearly dependent for all a
2) linearly independent for all a
3) linearly dependent for a < 0
4) linearly dependent for a > 0

25. If $\vec{x}$ and $\vec{y}$ be unit vectors and $|\vec{z}| = \frac{2}{\sqrt{7}}$ such that $\vec{z} + \vec{z}\times\vec{x} = \vec{y}$, then the angle θ between $\vec{x}$ and $\vec{z}$ can be

1) 30^0 2) 60^0 3) 90^0 4) None of these

26. If $z = \begin{bmatrix} O & I \\ I & O \end{bmatrix}$ where O, I are 2×2 null and identity matrix then det $(|z|)$ is

1) 1 2) -1 3) 0 4) None of these

27. Consider a circle, $x^2 + y^2 = 1$ and point $P\left(1, \sqrt{3}\right)$. PAB is a secant drawn from P intersecting circle in A and B (distinct) then range of [PA] + [PB] is

1) $\left[2, 2\sqrt{3}\right]$ 2) $\left(2\sqrt{3}, 4\right]$ 3) $(0,4]$ 4) None of these

28. Standard deviation of two distributions are same and their arithmetic means are 30 and 25 respectively. If their coefficient of variation is v_1 and v_2, then $6v_1 - 5v_2$ is equal to

1) 0 2) 1 3) 2 4) 3

29. Which of the following is true for a relation R : A $\rightarrow$ A

1) If R is reflexive then domain of R is A
2) If R is symmetric then domain of R is A
3) If R be an increasing function, then R is transitive
4) If R is one one function then R is symmetric

30. Rohith makes a statements p, q about future. if it is known that $p \to q$ and probability of occurance of p is $\frac{3}{5}$. What is the probability that q doesn't happen

1) $\frac{1}{5}$ 2) $\frac{2}{5}$

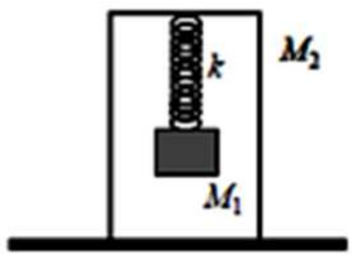
3) $\frac{3}{5}$ 4) can't be determined by given information

PHYSICS

31. The K, L and M energy levels of platinum lie roughly at 78, 12 and 3 keV respectively. The ratio of wavelength of K_α line to that of K_β line in X-ray spectrum is:

1) $\frac{22}{3}$ 2) $\frac{3}{22}$ 3) $\frac{22}{25}$ 4) $\frac{25}{22}$

32. A block of mass M_1 is hanged by a light spring of force constant k to the top bar of a reverse U-frame of mass M_2 on the floor. The block is pulled down from its equilibrium position by a distance x and then released. The minimum value of x such that the reverse U-frame will leave the floor momentarily, is:-



1) $x = (M_1 + M_2)g/k$ 2) $x = (2M_1 + M_2)g/k$

3) $x = (M_1 + 2M_2)g/k$ 4) $x = M_1 g/k$

33. Two waveform, $\psi_1(x,t) = A\sin(kx - \omega t)$ and $\psi_2(x,t) = A\sin(kx + \omega t + \pi/3)$, travelling along the x-axis, are superposed. The position of the nodes is given by $\left(k = \frac{2\pi}{\lambda}, \omega = 2\pi f\right)$

1) $x_n = \left(n + \frac{1}{6}\right)\frac{\lambda}{2}$ 2) $x_n = \left(n - \frac{1}{6}\right)\frac{\lambda}{2}$

3) $x_n = n\frac{\lambda}{2}$ 4) no nodes are formed

34. The main difference in the phenomenon of interference and diffraction is that :-

1) diffraction is due to interaction of light from the same wavefront whereas interference is the interaction of (light) waves from two isolated sources.

2) diffraction is due to interaction of light from same wavefront, whereas the interference is the interaction of two waves derived from the same source.

3) diffraction is due to interaction of waves derived from the same source, whereas the interference is the bending of light from the same wavefront.

4) diffraction is caused by the reflected waves from a source whereas interference is caused due to refraction of waves from a source.

35. A plastic circular disc of radius R is placed on a thin oil film, spread over a flat horizontal surface. The torque required to spin the disc about its central vertical axis with a constant angular velocity is proportional to :-

1) R^2 2) R^3 3) R^4 4) R^6

36. When the gap is closed without placing any object in the screw gauge whose least count is 0.005 mm, the 5th division on its circular scale coincides with the reference line on main scale, and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances by five times to the corresponding reading when no object was placed. There are 200 divisions on the circular scale. The radius of the sphere is :-

1) 4.10 mm 2) 4.05 mm 3) 2.10 mm 4) 2.05 mm

37. An engineer is given a fixed volume V_m of metal with which to construct a spherical pressure vessel. Interestingly, assuming the vessel has thin walls and is always pressurized to near its bursting point, the amount of gas the vessel can contain, n (measured in moles), does not depend on the radius r of the vessel; instead it depends only on V_m , the ideal gas constant R (measured in J/(K.mol)), and the tensile strength of the metal σ (measured in N/m^2). Which of the following gives n in terms of these parameters? (ignore atmospheric pressure)

1) $n=\frac{2}{3}\frac{V_m\sigma}{RT}$ 2) $n=\frac{2}{3}\frac{\sqrt[3]{V_m\sigma}}{RT}$ 3) $n=\frac{2}{3}\frac{\sqrt[3]{V_m\sigma^2}}{RT}$ 4) $n=\frac{2}{3}\frac{\sqrt[3]{V_m^2\sigma}}{RT}$

38. Imagine a planet whose diameter and mass are one half of those of the earth. The day's temperature of this planet reaches up to 800K. Escape velocity on the surface of the earth is 11.2 km/sec, $k=1.38\times10^{-23}\,J/K$. Consider that the molecules possess average kinetic energy. Among the following, choose the wrong statement

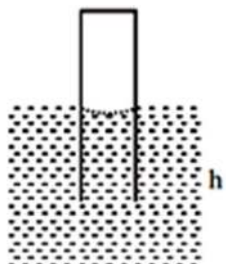
1) Oxygen molecules escape from the planet

2) Nitrogen molecules cannot escape from the planet

3) Argon molecules cannot escape from planet.

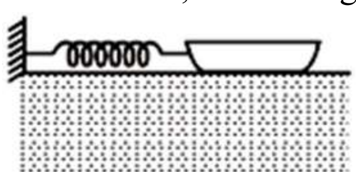
4) All the above is wrong

39. A glass capillary of length ℓ and inside radius r($r << \ell$) with its upper end sealed is introduced vertically into water. The atmospheric pressure is p_0. To what length h has the capillary to be submerged to make the water levels inside and outside the capillary coincide. Assume that temperature of air in the capillary remains constant. (Given, surface tension of water = T, angle of contact between glass water interface = $0°$)



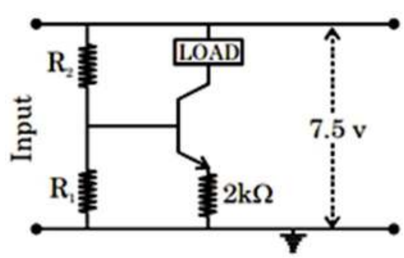
1) $\dfrac{\ell}{1+\dfrac{p_0 r}{T}}$ 2) $\dfrac{\ell}{1+\dfrac{p_0 r}{2T}}$ 3) $\dfrac{\ell}{1+\dfrac{p_0 r}{4T}}$ 4) $\dfrac{\ell}{1+\dfrac{2p_0 r}{T}}$

40. A steady force of 120 N is required to push a boat of mass 700 kg though water at a constant speed of 1 m/s. If the boat is fastened by a spring and held at 2m from the equilibrium position by a force of 450 N,find the angular frequency of damped SHM:



1) 0.56 rad/s 2) 0.21 rad/s 3) 1.35 rad/s 4) Motion is overdamped

41. In the transistor circuit shown, assume that the voltage drop between the base and the emitter is 0.5 V. What will be the ratio of the voltage across resistances R_2 & R_1 in order to make this circuit function as a source of constant current, I = 1mA? (Take $\alpha \to 1$)



1) 4.5 2) 3.0 3) 2.5 4) 2.0

42. Which of the following is heavily doped?

(a) Photo diode (b) Light emitting diode

(c) Zener diode (d) Solar cell

1) All four 2) (a), (b) & (c) 3) (b) & (c) 4) c only

43. A crew of scientists has built a new space station. The space station is shaped like a wheel of radius R, with essentially all its mass M at the rim. When the crew arrives, the station will be set rotating at a rate that causes an object at the rim to have radial acceleration g, thereby simulating Earth's surface gravity. This is accomplished by two small rockets, each with thrust T newtons, mounted on the sation's rim. How long a time t does one need to fire the rockets to achieve the desired condition ?

1) $t=\frac{\sqrt{gR^3}M}{(2T)}$ 2) $t=\frac{\sqrt{gR}M}{(2T)}$ 3) $t=\sqrt{\frac{gR}{\pi}}\frac{M}{T}$ 4) $t=\frac{\sqrt{gR}M}{(\pi T)}$

44. In the experiment of Davisson – Germer an electron beam of wavelength $8\overset{0}{A}$ falls normally on a crystal of unknown metal for which atomic spacing in the surface layers is $5\overset{0}{A}$. The scattering angle at which the first order maximum obtained is:-

1) $0°$ 2) $37°$ 3) $53°$ 4) $74°$

45. A scientist claims to have a perfect technique in which he can spontaneously convert an electron completely into energy in the laboratory without any other material required. What is the conclusion about this claim from our current understanding of physics?

1) This is possible because Einstein's equation says that mass and energy are equivalent…. It is just very difficult to achieve with electrons

2) This is possible and it is done all the time in the high-energy physics labs.

3) This is possible and it is done all the time in the low-energy physics labs.

4) This is not possible because charge conservation would be violated.

46. A long solenoid has 1000 turns per metre and carries a current of 1A. It has a soft iron core of $\mu_T = 1000$. The core is heated beyond the curie temperature, T_C.

1) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically.

2) The H and B fields in the solenoid are nearly unchanged.

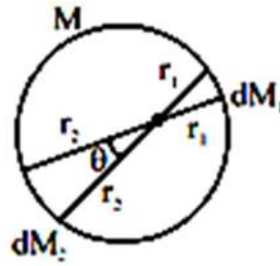
3) The magnetization in the core reverses direction.

4) The magnetization in the core increases by a factor of about 10^8

47. In the fuse of a diwali bomb the flame propagates with constant velocity $v_f = 0.8\,cm/sec$. What is the minimum length of the fuse a man should have so that he can run off to a safe distances s=120 m after igniting the flame, while the flame will not reach the bomb? The speed of person is $v_p = 4m/sec$.

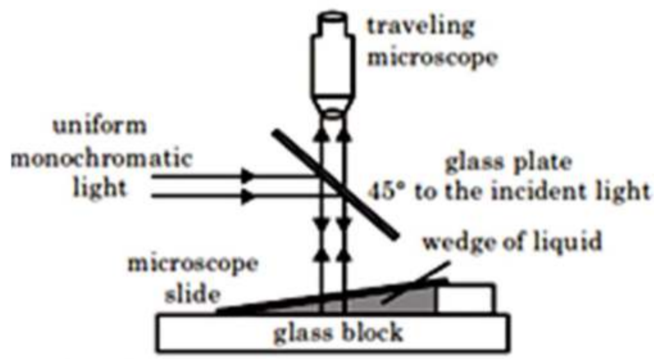
1) 24 m 2) 24 cm 3) 12 cm 4) 12m

48. Consider a uniform shell of mass M and radius R. Suppose a point mass m is placed inside shell as shown. Point mass lies at distance r_1 and r_2 from two opposite surfaces shown. Suppose dM_1 and dM_2 be the masses of surface elements on the two sides and they attract the particle with forces dF_1 and dF_2 respectively. Then which one of the following is true.**(Consider θ to be infinitesimal small solid angle):**



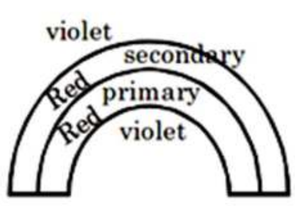
1) $\frac{dF_1}{dF_2}=\frac{r_2^2}{r_1^2}$ 2) $\frac{dF_1}{dF_2}=\frac{r_2}{r_1}$ 3) $dF_1 = dF_2$ 4) $\frac{dF_1}{dF_2}=\frac{r_1^2}{r_2^2}$

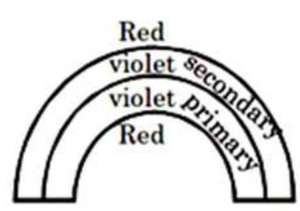
49. The following diagram is an arrangement to view the interference pattern produced by a very small angled wedge of liquid between a microscope slide and a glass block. The interference pattern obtained is made of equally spaced (or fringe width) may be decreased by some of the following actions (Consider one change at a time)

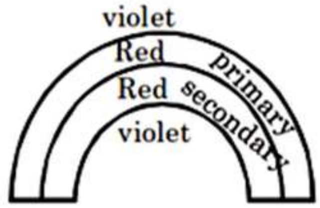


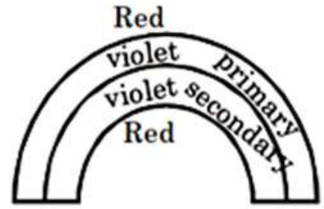
1) by increasing the angle of the liquid wedge.
2) by using a liquid of smaller refractive index.
3) by using a thicker glass block
4) by using a longer liquid wedge of the same angle.

50. In a rainbow, we see two rainbows; primary and secondary. Which of the following figures approximately represent the order of colors in rainbows?

1) 

2) 

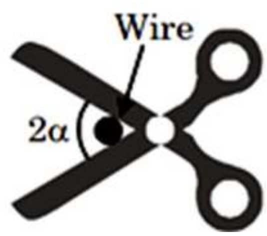
3) 

4) 

51. A simple telescope consisting of an objective of focal length 60 cm and a single eye lens of focal length 5 cm is focused on a distant object in such a way that parallel rays emerge from the eye lens. If the object subtend an angle of $2°$ at the objective then the angular width of the image is:-

1) $10°$ 2) $24°$ 3) $50°$ 4) $36°$

52. Some one is using a scissors to cut a wire of circular cross section and negligible weight. The wire slides in the direction away from the hinge until the angle between the scissors blades becomes 2α. The friction coefficient between the blades and the wire, is:-



1) $\sqrt{1-\tan\alpha}$ 2) $2\cos\alpha$ 3) $\tan\alpha$ 4) $2\tan\alpha$

53. Government has divided the carrier frequency range from 30 Mhz to 60 Mhz between 20 stations which are independently beaming their signals. Which of the signals can be beamed by each of the stations:-

1) Voice signals only
2) Music signals and voice signals only
3) TV Signals, Music Signals as well as voice signals
4) Music signals only

54. Figure (a) shows plot of voltage across the capacitor as a function of the driving frequency for a sinusoidally driven electromagnetic oscillator LCR circuit. Figure (b) shows phase angle ϕ (phase difference between voltage and current) vs ω/ω_0 graph for same circuit, for three different quality factor graph 1, 2, 3, of figure (a) and each one can be matched by one of graphs a, b, c also of figure (b):-

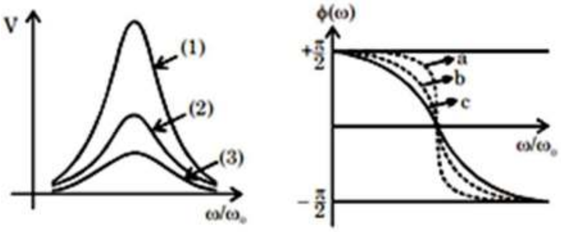
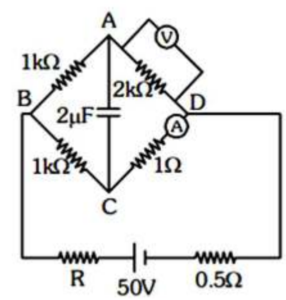


figure (a)　　　　figure (b)

1) Graph (3) corresponds to graph (a)
2) Graph (1) corresponds to graph (c)
3) The circuit of graph 1 has high quality factor.
4) The circuit of graph 3 has high quality factor.

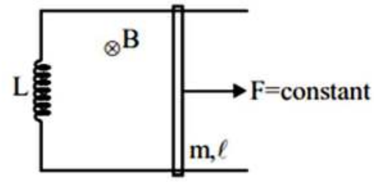
55. Negative charge is revolving around a fixed positive charge in a circular orbit. If the classical idea of an accelerating charge radiating energy is valid, then the negative charge will not:
1) spiral towards the positive charge, with increasing kinetic energy
2) spiral towards the positive charge with potential energy decreasing at a faster rate than increase in its kinetic energy.
3) spiral away from the positive charge and finally escape from the binding of the positive charge
4) revolve around the positive charge with increasing frequency of revolution.

56. Calculate the energy stored in the capacitor of capacitance $2\mu F$, at the instant when the voltmeter 'V' gives a reading of 15V and the ammeter A reads 15 mA. Resistance of voltmeter is unknown and ammeter is 999Ω.



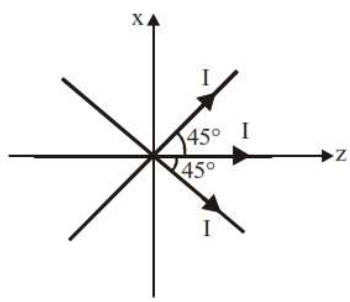
1) $5\mu J$　　2) $10\mu J$　　3) $0.5\mu J$　　4) 0

57. A loop is formed by two parallel conductors conducted by a solenoid with inductance L and a conducting rod of mass m, which can freely (without friction) slide over the conductors. The conductors are located in a horizontal plane in a uniform vertical magnetic field with induction B. The distance between the conductors is equal to ℓ. At the moment $t = 0$ a constant force F starts acting on the rod. Angular frequency of motion rod is ω. If F and m are doubled then:



1) ω become $\frac{1}{\sqrt{2}}$ times
2) ω become $\sqrt{2}$ times
3) ω become 2 times
4) ω become $\frac{1}{2}$ times

58. Three long straight wires in the XZ-plane, each carrying I, cross at the origin of coordinates, as shown in the figure below. Let $\hat{x}, \hat{y}$ and $\hat{z}$ denote the unit vectors in the x,y and z-directions respectively. The magnetic field B as a function of x, with $y = 0$ and $z = 0$,is:



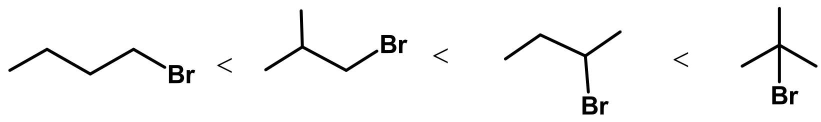
1) $B = \frac{3\mu_0 I}{2\pi x}\hat{x}$
2) $B = \frac{3\mu_0 I}{2\pi x}\hat{y}$
3) $B = \frac{\mu_0 I}{2\pi x}\left(1+2\sqrt{2}\right)\hat{y}$
4) $B = \frac{\mu_0 I}{2\pi x}\left(1+2\sqrt{2}\right)-\hat{y}$

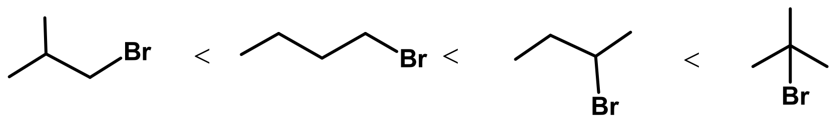
59. The current voltage relation of diode is given by $I = \left(e^{1000V/T}-1\right)$ mA, where the applied voltage V is in volt and the temperature T is in kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300K, what will be the error in the value of current in mA?
1) 0.2 mA 2) 0.02 mA 3) 0.5 mA 4) 0.05 mA

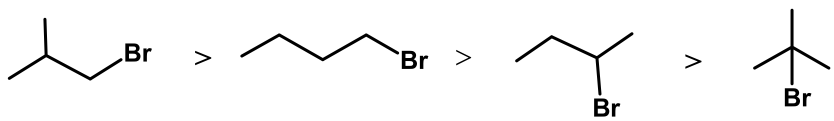
60. A pendulum clock loses 12 s a day if the temperature is $40°C$ and gains 4 s in a day if the temperature is $20°C$. The temperature at which the clock will show correct time, and the coefficient of linear expansion (α) of the metal of the pendulum shaft are, respectively
1) $25°C, \alpha = 1.85\times10^{-5}/°C$
2) $60°C, \alpha = 1.85\times10^{-4}/°C$
3) $30°C, \alpha = 1.85\times10^{-3}/°C$
4) $55°C, \alpha = 1.85\times10^{-2}/°C$

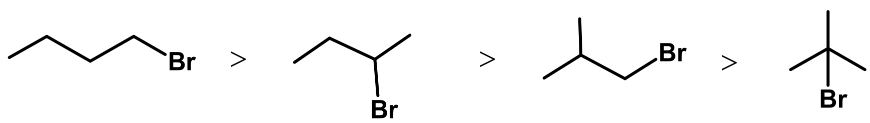
CHEMISTRY

61. Which of the following is correct regarding nucleophilic substitution reaction?

1)  (S_N1)

2)  (S_N1)

3)  (S_N2)

4)  (S_N2)

62. In an excited He^+ ion an electron falls from 3rd excited state to the 1st excited state emitting a photon which is suitable for excitation of an electron in a H atom from $(n-1)^{th}$ to n^{th} shell. The value of n is

1) 2 2) 3 3) 4 4) 5

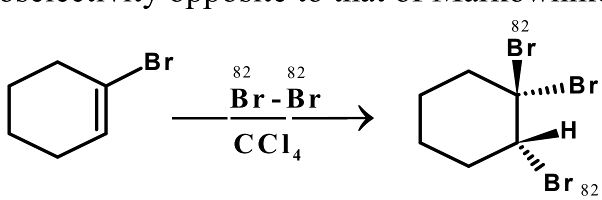
63. Which of the following complex will not show geometrical isomerism?

1) $\left[Pt(NH_3)_2ClNO_2\right]$ 2) $\left[Pt(NH_3)_2Cl_2\right]$

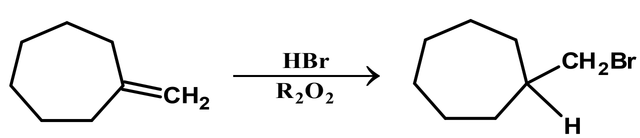
3) $[Co(en)_3]^{3+}$ 4) $\left[Co(NH_3)_3Cl_3\right]$

64. Mention True(T) and false (F) out of the following statements:

S_1 :In hydroboration oxidation of alkene, H and OH are introduced with a regioselectivity opposite to that of Markownikove's rule.



S_2: $\left(\overset{82}{Br}\text{ is radioactive isotope of bromine}\right)$



methylene cycloheptane bromomethyl cycloheptane

S_3 :

Codes:

1) T T T 2) T T F 3) T F T 4) F T F

65. EMF diagram for Mn is given as

(species involved are MnO_4^- , MnO_4^{-2} , MnO_2 , Mn^{+3} , Mn^{+2} , Mn^0)

$$MnO_4^- \xrightarrow{+0.56V} MnO_4^{2-} \xrightarrow{+2.26V} MnO_2 \xrightarrow{+0.95V} Mn^{3+} \xrightarrow{+0.51V} Mn^{2+} \xrightarrow{-1.81V} Mn^0$$

$$MnO_4^- \xrightarrow{+1.7V} MnO_2 \qquad MnO_2 \xrightarrow{+1.23V} Mn^{2+}$$

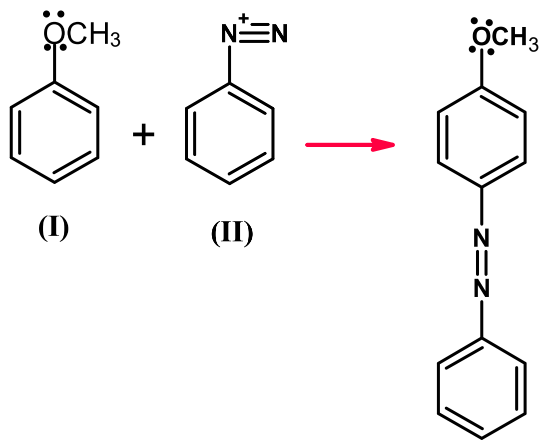
E^0 value for $MnO_4^- - Mn^{3+}$ couple is?

1) 1.5 v 2) 1.05 v 3) 2.45 v 4) 0.45 v

66. $[Fe(CN)_6]^{3-}$ ion has magnetic moment of 1.73 BM then $[Fe(H_2O)_6]^{3+}$ has a magnetic moment of

1) 1.73 BM 2) > 1.73 BM 3) < 1.73 BM 4) 0 BM

67. For the given reaction consider the statement 1, 2 and 3



(I) (II)

1) Presence – of - NO_2 group in aromatic ring **(I),** will increase the rate of reaction.

2) Presence of - NO_2 group in aromatic ring **(II),** will increase the rate of reaction.

3) In this reaction a wheland intermediate (σ complex) is formed.

1) T T T 2) F F F 3) F T T 4) F T F

68. For three different gases values of Vander waal's constants 'a' and 'b' are given. What is the order of liquefaction of gases:

Gases	a	b
X_2	1.3	0.09
Y_2	4.1	0.023
Z_2	2.2	0.075

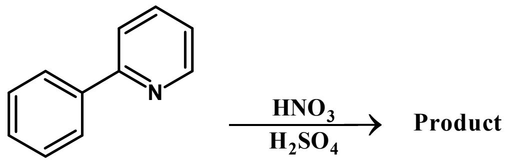
1) $X_2 > Y_2 > Z_2$ 2) $Y_2 > Z_2 > X_2$ 3) $Z_2 > Y_2 > X_2$ 4) $X_2 > Z_2 > Y_2$

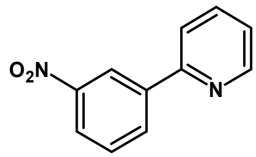
69. What is the hybridization of Boron and nitrogen in Borazine?

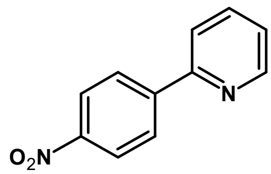
1) Both sp^3 2) N is sp^3 and B is sp^2 hybridised

3) Both sp^2 4) N is sp^2 and B is sp^2 hybridised

70. Which of the following statement is/are correct about the product



1) The product is 

2) The product is 

3) Pyridine acts as a deactivating and O, p-directing

4) Pyridine acts as a activating and meta directing

71. The specific conductivity of 0.001 M solution of certain monoprotic acid is 4.95×10^{-5} s cm^{-1}. If λ^0 for the acid is 495 S cm^2 mol^{-1} than its degree of dissociation & dissociation constant respectively are

1) 0.01 & 1.1×10^{-6} 2) 0.1 & 1.1×10^{-5}

3) 0.01 & 10^{-8} 4) 0.01 & 1.1×10^{-9}

72. Which of the following chloride does not respond to the chromyl chloride test?

1) NaCl 2) $CaCl_2$ 3) $AlCl_3$ 4) Hg_2Cl_2

73. For the given reaction choose the correct option:

$$\underset{\displaystyle Br}{CH_3-CH_2-\underset{|}{CH}-CH_3} \xrightarrow{\text{alc.KOH}/\Delta}$$

1) Reaction follows $\mathbf{E_2}$ pathway 2) Reaction follows $\mathbf{E_1}$ pathway

3) Reaction follows $\mathbf{S_N1}$ pathway 4) Reaction follows $\mathbf{S_N2}$ pathway

74. A mineral having the formula AB_2 crystallizes in the cubic close-packed lattice, with the A atoms occupying the lattice points. If all the atoms along the body diagonal are removed the formula of the resultant crystal is

1) $A_{30}B_{48}$ 2) $A_{28}B_{46}$ 3) $A_{35}B_6$ 4) $A_{37}B_7$

75. Formation of metallic copper from the sulphide ore in the commercial metallurgical process involves.

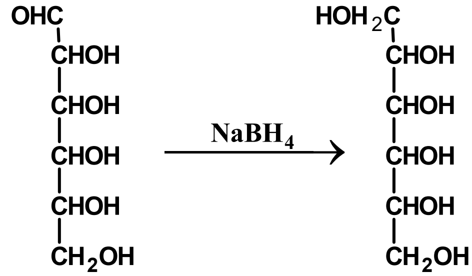
1) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $Cu_2O + C \rightarrow 2Cu + CO$

2) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $2Cu_2O + SO_2 \rightarrow 3Cu + CuO + SO_3$

3) $Cu_2S + 2O_2 \rightarrow CuSO_4$; $CuSO_4 + Cu_2S \rightarrow 3Cu + 2SO_3$

4) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $Cu_2O + CO \rightarrow 2Cu + CO_2$

76. Observe the following reaction and find out that how many number of reactant stereoisomers can be reduced to optically inactive meso products



1) 2 2) 4 3) 6 4) 8

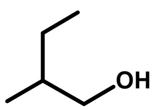
77. The vapour pressure of benzene, Toluene and Xylene are 75 torr, 22 torr and 10 torr at 20ºC. Which of the following is not a possible value of the vapour pressure of an equimolar binary/ternary solution of these at 20ºC? Take all solutions as ideal in nature (give approximate answer)

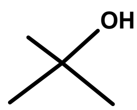
1) 48 2) 16 3) 35 4) 53

78. Acidified solution of potassium dichromate on treatment with H_2O_2 in the presence of ether yields

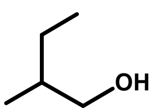
1) $CrO_3 + H_2O + O_2$ 2) $Cr_2O_2 + H_2O + O_2$
3) $CrO_5 + H_2O + K_2SO_4$ 4) $H_2Cr_2O_7 + H_2O + O_2$

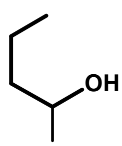
79. Which one of the following match is correct

1) 

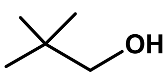
& 

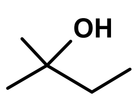
are position isomers

2) 

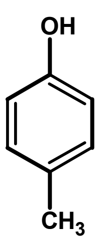
& 

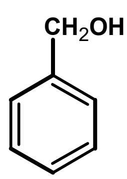
are position isomers

3) 

& 

are chain isomers

4) 

& 

are Homologous

80. The equilibrium constant for the reaction

$$x^{2+} + y^{4+} \rightleftharpoons X^{3+} + y^{3+}$$

$E^0_{y^{4+}/y^{3+}} = 1.44V$

$E^0_{x^{3+}/x^{2+}} = 0.85V$ is

1) 10^{10} 2) 10^{12} 3) 2×10^9 4) 2×10^{10}

81. Lanthanum is grouped with *f*-block elements because

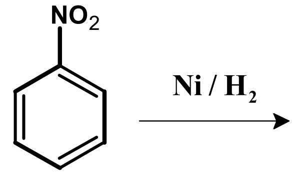
1) It has partially filled *f* -orbitals

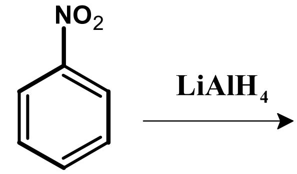
2) It is just before *Ce* in the periodic table

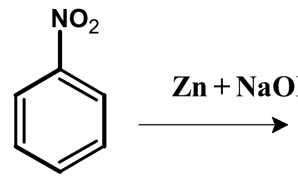
3) It has both partially filled *f* and *d*-orbitals

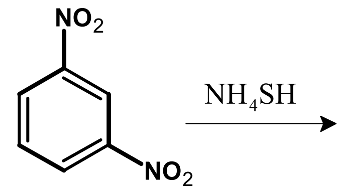
4) The properties of Lanthanum are very similar to the elements of 4*f* block

82. Identify reaction which gives aniline as a product:-

1) 

2) 

3) 

4) 

83. One mole ideal gas expands isothermally at 27^0C into an evacuated vessel so that pressure drops from 1.01×10^6 N m^{-2} to 1.01×10^5 N m^{-2}. Then

(1) ΔS_{system} is 19.15 J / K.mol (2) $\Delta S_{surrounding}$ is zero

(3) $\Delta S_{surrounding}$ is -19.15 J/K.mol (4) $\Delta S_{universe}$ is zero

(5) $\Delta S_{universe}$ is 19.15 J/K mol

1) 1,3 & 4 **2)** 1, 2 & 5 3) only 4 4) only 5

84. $Br_2 + OH^-(Hot) \rightarrow (P) + (Q); (P) + (Q) + H^+ \rightarrow Br_2$; (P) gives yellow precipitate with $AgNO_3$, (P) and (Q) are

1) Br^-, BrO^- 2) Br^-, BrO_3^- 3) BrO^-, BrO_3^- 4) BrO^-, BrO_4^-

85. The monomers of Buna-S polymers are:

1) Vinyl chloride and Vinylidene 2) Styrene and butadiene

3) Acrylonitrile and butadiene 4) Isobutylene and isoprene

86. If at a certain temperature, the following equilibrium is established,

$$CO(g) + NO_2(g) \rightleftharpoons CO_2(g) + NO(g)$$

One mole of each of the four gases is mixed in one litre container and the reaction is allowed to reach equilibrium state. When excess of baryta water is added to the equilibrium mixture, the weight of white precipitate obtained is 236.4 g. The equilibrium constant, K_c of the reaction is (at.wt of Ba=137)

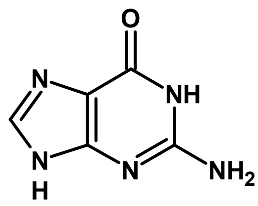
1) 1.2 2) 2.25 3) 2.1 4) 3.6

87. In which of the following S – S link is present?

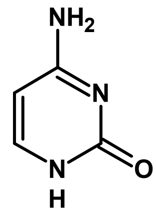
1) Caro's acid 2) Dithionic acid

3) pyrosulphuric acid 4)sulphuric acid

88.



Guanine



Cytosine

In a DNA double - helix, guanine and cytosine bases, shown above, are paired together by

1) covalent bonds
2) hydrogen bonds
3) peptide bonds
4) Ionic bond

89. Equal masses of $KMnO_4$ (M.W. = 158) and $K_2Cr_2O_7$ (M.W. = 294) are treated with excess of KI in acidic medium. In which case more amount of Iodine is liberated?

1) By $KMnO_4$
2) By $K_2Cr_2O_7$
3) Same in both cases
4) $K_2Cr_2O_7$ cannot oxidize KI in acidic medium

90. Which of the following reactions are disproportionation reactions?

$(a)\, Cu^{+} \rightarrow Cu^{+2} + Cu$

$(b)\, 3MnO_4^{-2} + 4H^{+} \rightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$

$(c)\, 2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

$(d)\, 2MnO_4^{-} + 3Mn^{+2} + 2H_2O \rightarrow 5MnO_2 + 4H^{+}$

1)a,b
2) a,b,c
3) b,c,d
4) a,d

KEY SHEET

1	**3**	2	**1**	3	**1**	4	**3**	5	**2**
6	**4**	7	**2**	8	**2**	9	**2**	10	**3**
11	**4**	12	**2**	13	**4**	14	**3**	15	**2**
16	**1**	17	**3**	18	**1**	19	**2**	20	**2**
21	**1**	22	**2**	23	**1**	24	**2**	25	**2**
26	**2**	27	**2**	28	**1**	29	**1**	30	**4**
31	**4**	32	**1**	33	**2**	34	**2**	35	**3**
36	**4**	37	**1**	38	**1**	39	**2**	40	**1**
41	**4**	42	**3**	43	**2**	44	**4**	45	**4**
46	**1**	47	**2**	48	**3**	49	**1**	50	**1**
51	**2**	52	**3**	53	**2**	54	**3**	55	**3**
56	**4**	57	**1**	58	**3**	59	**1**	60	**1**
61	**1**	62	**1**	63	**3**	64	**1**	65	**1**
66	**2**	67	**3**	68	**2**	69	**3**	70	**1**
71	**2**	72	**4**	73	**1**	74	**1**	75	**1**
76	**2**	77	**4**	78	**3**	79	**3**	80	**1**
81	**4**	82	**1**	83	**2**	84	**2**	85	**2**
86	**2**	87	**2**	88	**2**	89	**1**	90	**1**